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CSC360 - Computer Graphics and Digital Image Processing
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Reflection: Computer Graphics & Digital Image Processing

The practical session on Digital Image Processing and Computer Graphics (CSC360) enabled me to learn how the concepts of computer graphics could be applied using Java. This was important since it helped to relate theory of computer graphics concepts such as coordinates, geometric representations, object-oriented programming, and graphics to a practical program.

The primary assignment in this practical was to develop the Square Project in Java using javax.swing and java.awt libraries. In this case, the creation of the square did not require using the pre-defined rectangle or square. The square was made up of four line segments. The project was created by extending the JPanel class and overriding the paintComponent() method to implement the drawing process. Graphics2D class performs the rendering process whereas BasicStroke determines the thickness of the line. I learned about the concept of anti-aliasing, which smoothens the line making it more aesthetically pleasing. A JFrame is the window where the square will be displayed.

One key thing I learned during the unit is that we could use mathematical formulas to establish the coordinates of the square from its center point and side length. With the center being represented as C(Cx, Cy) and the length as l, the four coordinate points will be as follows: (Cx-l/2, Cy-l/2), (Cx+l/2, Cy-l/2), (Cx-l/2, Cy+l/2), and (Cx+l/2, Cy+l/2). Having the center of the canvas being the default center makes the programming process easier since the square becomes independent of any other fixed coordinates. It helped me understand how the geometry of mathematics is transformed into graphical coordinates.

The lecture provided an insight into the structure of the Maven projects and the importance of the pom.xml file. It was learnt that the pom.xml file includes all the information of a particular project including group ID, artifact ID, version number and any dependencies needed. Also, it was made clear that the src/main folder is where the production source code is placed while the src/test folder is where the testing takes place and the target folder is where the generated build output is placed.

In addition to the above, I was able to understand the difference between an object field and a class field. While the former is unique to each object with each object having its own copy, the latter is unique to the class, with all objects sharing it. All in all, the practical helped me gain more knowledge in Java graphics, coordinate geometry, Maven projects, libraries, and object oriented programming.